Ignatius Nigel

nigelignatius505@gmail.com | +919751585709 | Tamil Nadu, India

Aerospace Engineering graduate with hands-on project experience across CAD design, CFD analysis, composite structures, and UAV systems. Has taken multiple designs from concept through fabrication and flight-testing, with results validated through simulation and physical testing. Quick to learn new tools and comfortable working independently or as part of a team.

Education

KCG College of Technology, Chennai - B.E in Aerospace Engineering Graduated 2026

Projects & Experience

Autonomous Satellite Collision Avoidance & Maneuver Optimization System Aug 2025-Mar 2026

- Developed a conjunction risk assessment framework for LEO satellites using TLE data and orbital propagation techniques.
- Implemented covariance-based uncertainty modeling for collision probability estimation.
- Optimized maneuver timing by ~8–12% for collision avoidance using Python and MATLAB simulations.

RC Aircraft Design & Payload Delivery Systems Nov 2023-Oct 2024

- Designed, modeled, fabricated, and tested fixed-wing UAVs capable of carrying payloads up to 5 kg.
- Created detailed CAD models and assemblies for manufacturing and testing.
- Integrated payload release mechanisms and optimized structural and aerodynamic performance.
- Improved flight stability and aerodynamic efficiency by 10–15% through iterative design validation and testing.

Supersonic Nozzle Flow Visualization & CFD Analysis Nov 2024-Jan 2025

- Performed CFD analysis of convergent-divergent nozzles at Mach 1.6, 1.8, and 2.0 using ANSYS Fluent. Evaluated shock-wave formation and expansion behavior.
- Validated pressure and velocity contours against theoretical gas-dynamics calculations.

Structural Optimization of Airframes using Composite Materials May 2024-Jul 2024

- Designed, modeled, and fabricated vacuum-bagged carbon fiber and fiberglass structures using engineering design principles.
- Achieved weight reduction by 15–20% while maintaining structural integrity.
- Applied composite material principles for high strength-to-weight ratio designs.

Publication

Satellite Collision Avoidance and Maneuver Optimization - 7th ICCII Conference

Published paper and received an award for presentation and contribution, along with certification.

Competitions

Technoxian WRC - RC Craft Challenge International

Demonstrated strong understanding of aerodynamics and stability optimization under competitive constraints.

HAC'24 - Aeromodelling Competition HITS, Chennai

Integrated payload delivery mechanisms with optimized flight control and structural efficiency.

Shaastra Aeromodelling Competition IIT Madras, Chennai

Focused on aerodynamic efficiency improvements through iterative design and testing.

Boeing Aeromodelling Competition IIT Madras, Chennai

Applied mission-based design strategies emphasizing payload integration and performance optimization.

Kurukshetra Aeromodelling Event Anna University, Chennai

Gained hands-on experience in aerodynamic analysis and real-time flight testing techniques.

Achievements & Recognition

- Consistently ranked as a finalist in multiple national and international aeromodelling competitions among 50+ teams, for aircraft stability and mission-based flight performance.
- Demonstrated the ability to transition from complex CAD designs to flight-ready airframes under strict timelines.

Skills

CAD & Simulation: SolidWorks, CATIA, Fusion 360, ANSYS Fluent (CFD), OpenRocket.
Engineering & Manufacturing: UAV Design, Composite Structures, Product Development, Propulsion Systems, 3D Printing.
Computing & Software: Python, MATLAB, Microsoft Office, Technical Documentation.